

KAKINADA INSTITUTE OF ENGINEERING AND TECHNOLOGY -II

DEPARTMENT OF CSE - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Animal Image Classification

Under the esteemed guidance of

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INTRODUCTION

Animal Image Classification is a computer vision task that uses deep learning models, such as EfficientNet, to automatically recognize and categorize images of animals into predefined species. EfficientNet, known for its high accuracy and computational efficiency, analyzes visual features like shape, color, texture, and patterns to distinguish between different animals. By leveraging such advanced architectures, this technology significantly enhances applications in wildlife monitoring, biodiversity research, pet identification, agriculture, and education, making it a powerful tool in both scientific and real-world settings.

ABSTRACT

This project presents an AI-based solution for classifying animals from images using deep learning. The model is built using EfficientNetB3, a state-of-the-art convolutional neural network known for its balance between accuracy and efficiency. The system is trained on a dataset containing images of 90 different animal species and achieves high classification accuracy. To make the model accessible and easy to use, a Streamlit-based web application is developed, allowing users to upload an image and instantly get the predicted animal name along with a confidence score. This solution has potential applications in wildlife monitoring, educational tools, and conservation efforts, providing a fast and reliable method for automated animal identification.

PURPOSE OF THE PROJECT

- Automate the identification of animals.
- Reduce manual work and time in image labeling.
- Useful for wildlife monitoring, education, and biodiversity tracking.
- Provide an efficient and scalable system.

Literature Survey

- VGGNet: Deep but slow and has many parameters.
- ResNet: Introduced skip connections, better accuracy.
- MobileNet: Lightweight but not always accurate.
- EfficientNet: Balances accuracy and size using compound scaling.
- We chose EfficientNetB3 for its excellent trade-off between performance and efficiency.

REGION OF INTRESET

1. **ROI is the part of the image where the animal is located.**
2. **Focusing on ROI removes background distractions** like trees, sky, or objects.
3. **Improves classification accuracy** by highlighting important features (shape, fur, face, etc.).
4. **Reduces image size and speeds up processing.**
5. **Can be extracted using object detection** (e.g., YOLO, Faster R-CNN).
6. **Essential for models to learn the correct features** without being confused by surroundings.



PROPOSED METHOD

The proposed method is a deep learning-based image classification system that identifies the species of an animal from a given image. The core of the system is the EfficientNetB3 model, which is a convolutional neural network pre-trained on ImageNet and fine-tuned for animal species classification.

STEPS IN PROPOSED METHOD

1. Image Input

- The user uploads an animal image through the web interface (Streamlit).

2. Preprocessing

- The image is resized to 224 X 224 pixels (required input size for EfficientNetB3).
- The pixel values are normalized (scaled between 0 and 1).

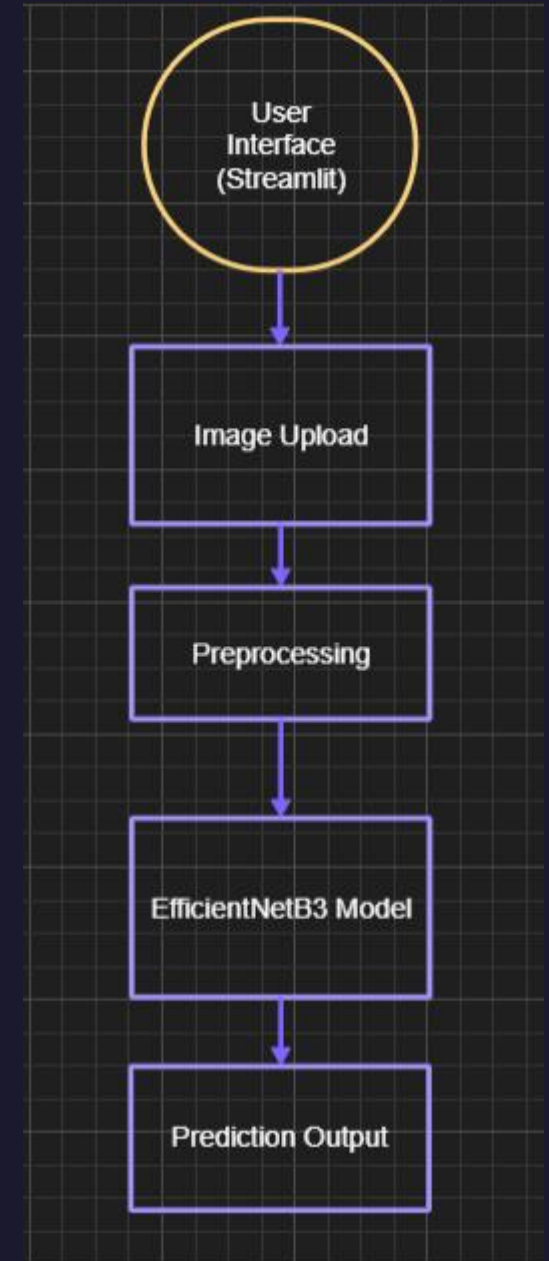
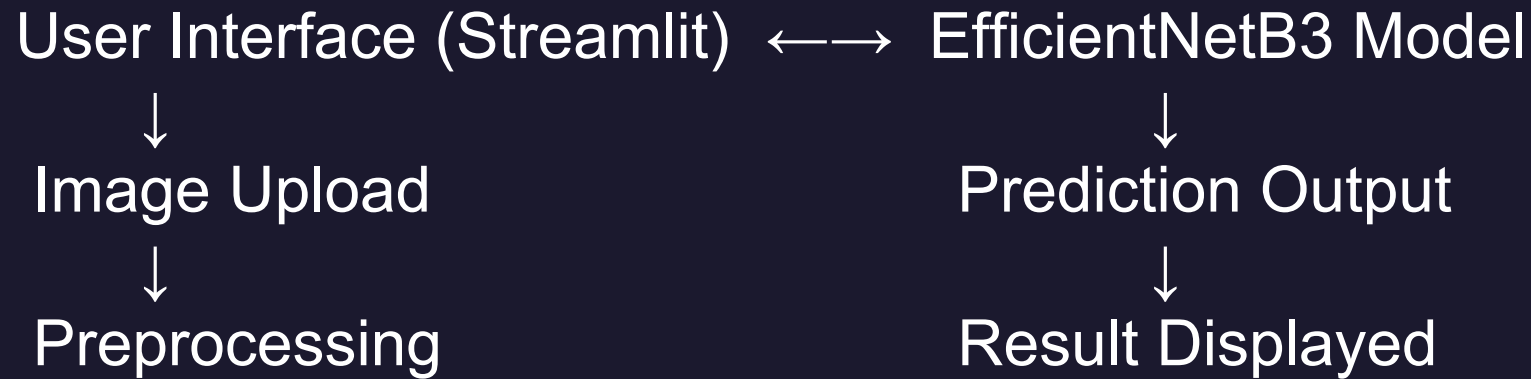
3. Feature Extraction & Classification

- The preprocessed image is passed through the EfficientNetB3 model, which extracts features and performs classification.
- The model outputs a list of possible animal species with confidence scores.

4. Result Display

- The top predicted animal name is displayed along with its confidence score on the web interface.

BLOCK DIAGRAM



ALGORITHM

EfficientNet is a powerful convolutional neural network (CNN) architecture that achieves high accuracy with fewer parameters and faster computation compared to traditional models. In the context of animal image classification, EfficientNetB3 proves to be an ideal choice due to its balanced scaling of network depth, width, and input resolution. This helps the model learn fine-grained features such as fur patterns, body shapes, and background context, which are essential for accurately distinguishing between various animal species.

By leveraging transfer learning from a model pre-trained on ImageNet and fine-tuning it with an animal-specific dataset, EfficientNetB3 provides a robust solution that generalizes well to new and unseen animal images, making it suitable for real-world applications like wildlife monitoring, species tracking, and educational tools.

EfficientNet model architecture

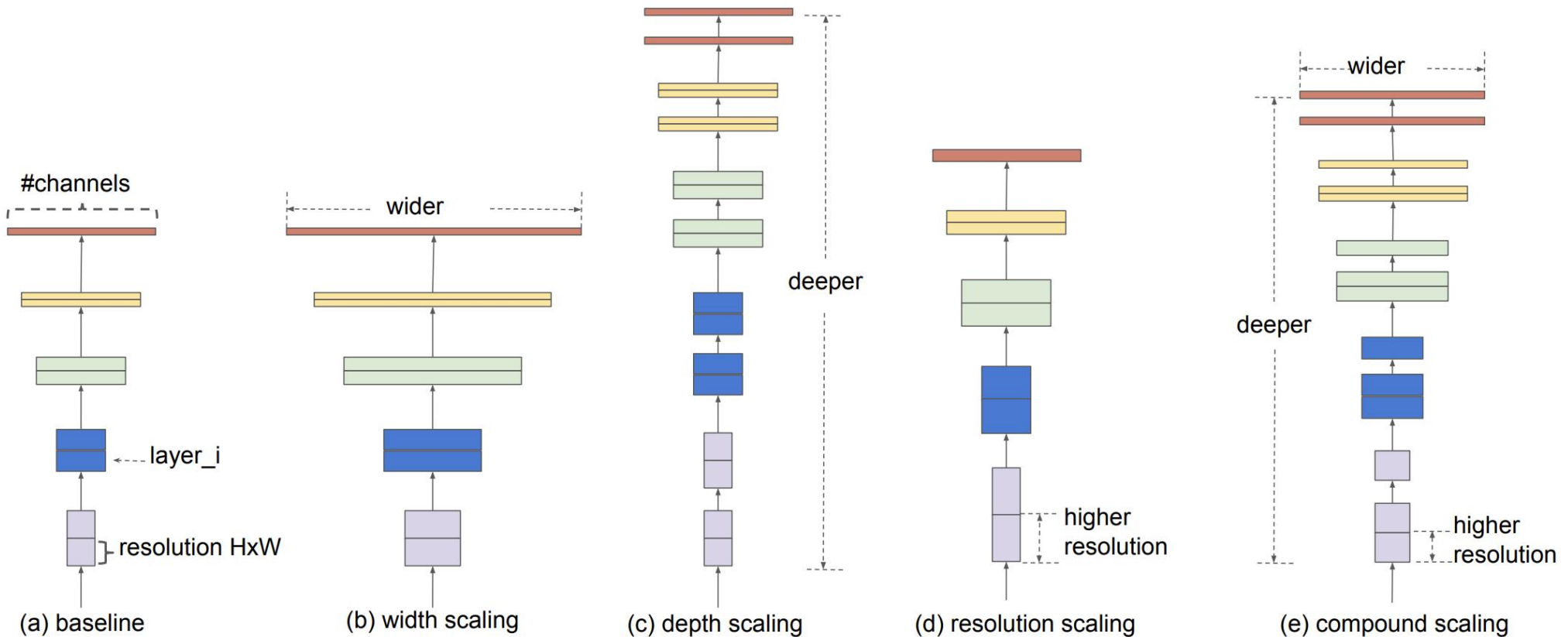
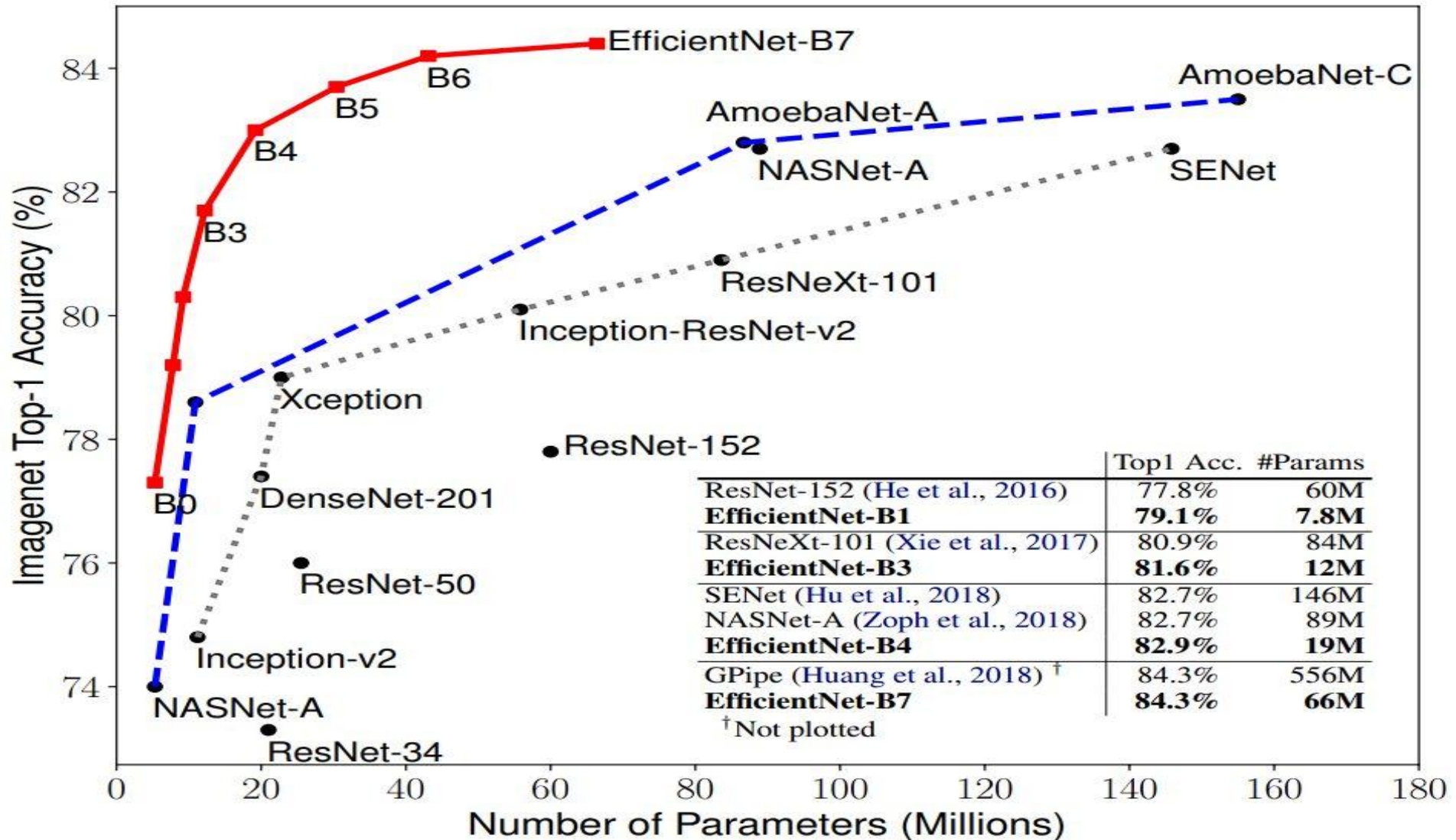


Figure 2. Model Scaling. (a) is a baseline network example; (b)-(d) are conventional scaling that only increases one dimension of network width, depth, or resolution. (e) is our proposed compound scaling method that uniformly scales all three dimensions with a fixed ratio.

Efficientnet model architecture Parameters and Accuracy



IMPLEMENTATION

1. Dataset Collection and Preprocessing

The dataset includes 90 labeled images of various animal species such as cats, dogs, lions, and elephants. Preprocessing steps involve resizing images to a fixed input size compatible with EfficientNetB3, normalizing pixel values, and splitting the data into training and testing sets. Optional steps like grayscale conversion can also be applied based on requirements.

2. Model Training with EfficientNetB3

EfficientNetB7, a high-performance deep learning model, is used for image classification. A pre-trained EfficientNetB3 model is fine-tuned on the animal dataset using transfer learning. The model training is carried out in a Kaggle Notebook with GPU support for faster computation. The best model weights are saved after training for future inference.

3. Animal Image Classification

The trained model is used to classify new animal images into their respective categories based on visual features like shape, texture, and color. This enables accurate and efficient species identification for practical applications in fields like education, wildlife conservation, and pet recognition.

SYSTEM ARCHITECTURE:

Hardware Requirements

- CPU: Multi-core (Intel i5/Ryzen 5 or higher)
- GPU: NVIDIA GTX 1060/RTX 2060 or similar (4GB+ VRAM)
- RAM: 8GB (16GB recommended)
- Storage: SSD (20GB+ free space)

Software Requirements

- OS: Linux (Ubuntu 18.04+), macOS, or Windows 10+
- Python: 3.7+
- Libraries: TensorFlow 2.x, PyTorch, EfficientNet, Pillow, NumPy, Matplotlib

Deep Learning Framework Compatibility

- TensorFlow/Keras: TensorFlow 2.x



Advantages

- Fast and Accurate Identification:

Quickly identifies animals in images with high accuracy, saving time compared to manual work.

- Supports Wildlife Conservation:

Helps monitor endangered species and detect illegal poaching activity in protected areas.

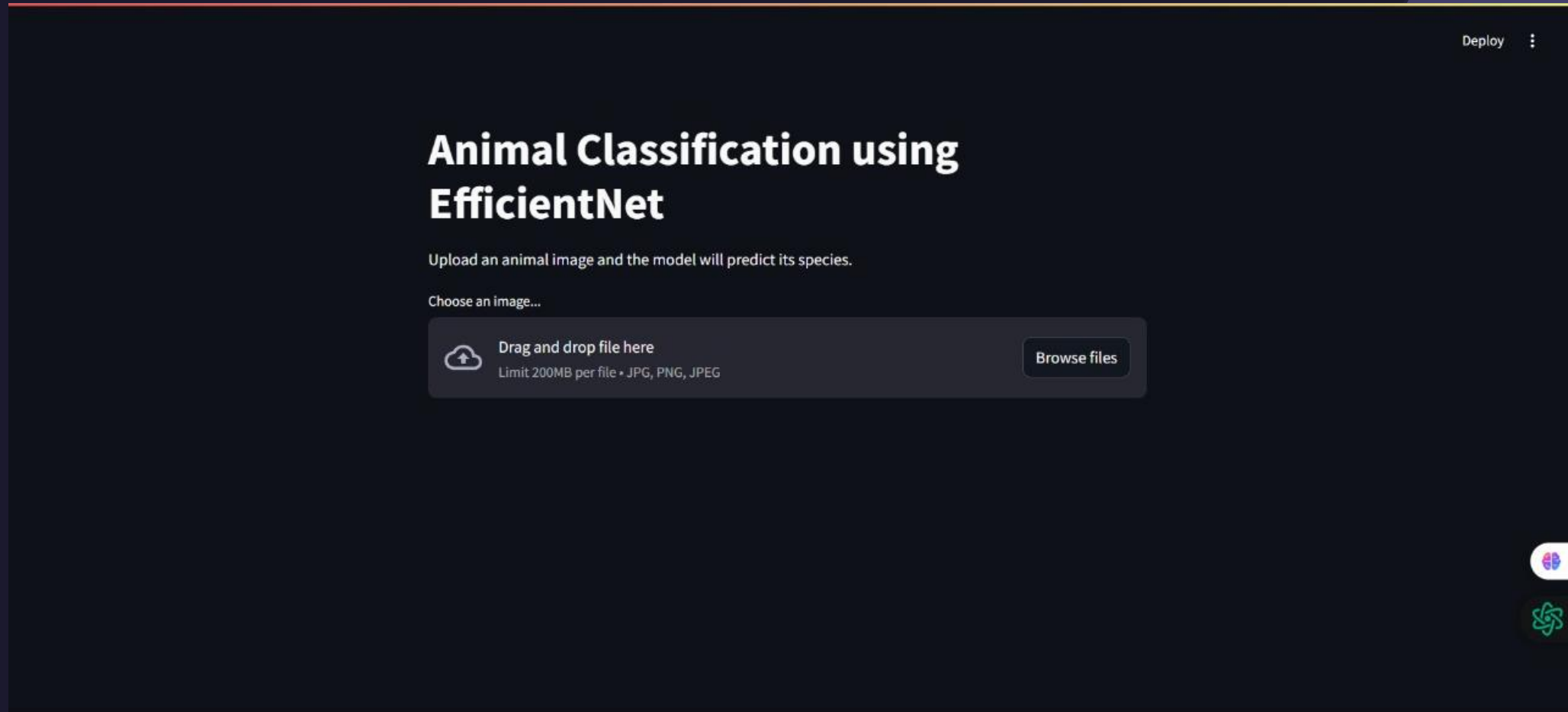
- Enhances Biodiversity Research:

Assists scientists in tracking animal populations and understanding species diversity.

- Automates Data Analysis:

Processes large amounts of images from camera traps or drones automatically.

USER INTERFACE



OUTPUT/RESULT

Animal Classification using EfficientNet

Upload an animal image and the model will predict its species.

Choose an image...



Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files



download.jpg 5.1KB



The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.



Uploaded Image

Prediction: lion (99.98%)

Animal Classification using EfficientNet

Upload an animal image and the model will predict its species.

Choose an image...



Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files



download.jpg 6.1KB



The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.



Uploaded Image

Prediction: cat (99.99%)

FUTURE SCOPE

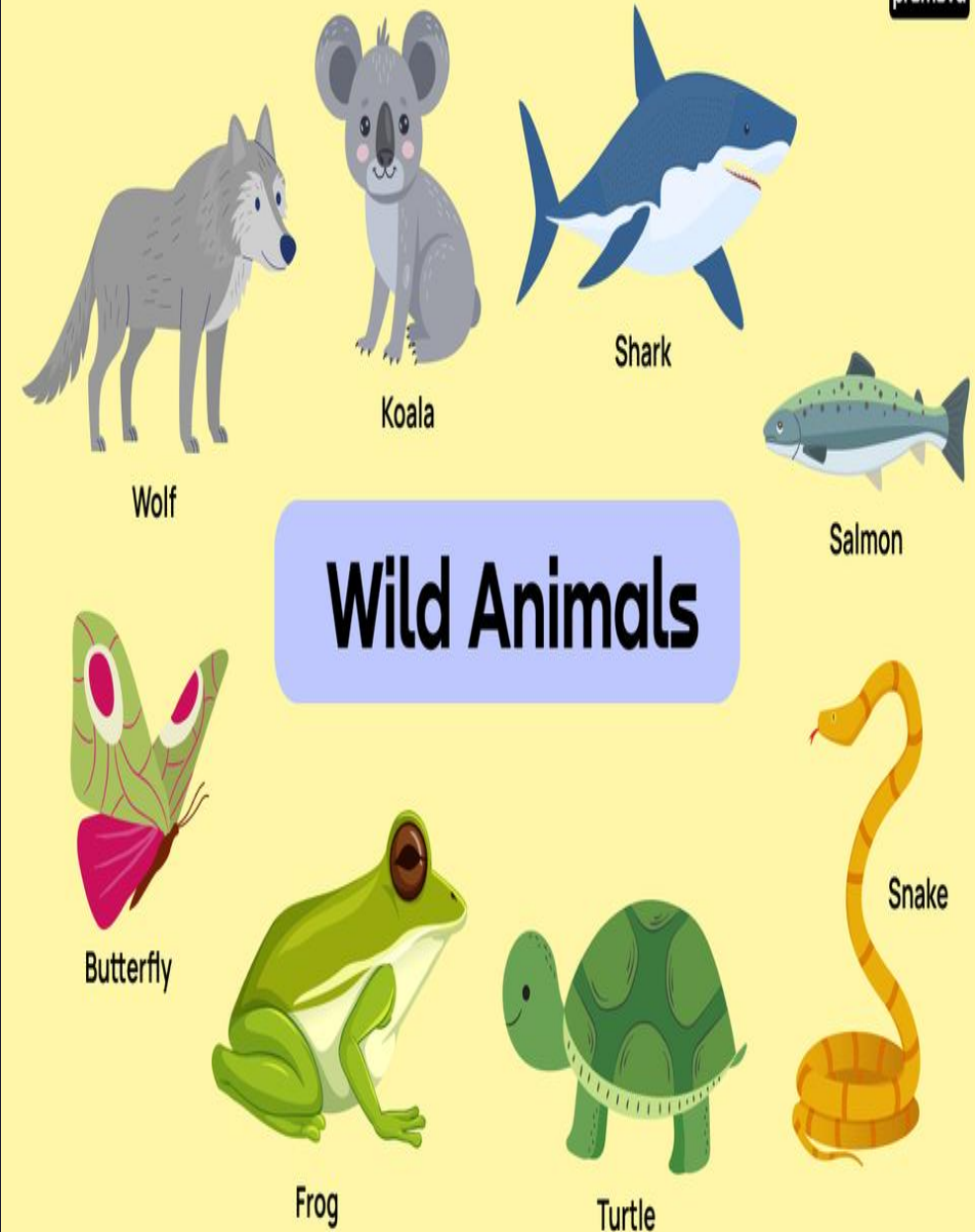
1. Wildlife Conservation: Automate animal detection in camera traps to study behavior, migration, and populations.
2. Agriculture & Livestock: Monitor livestock health, detect diseases, and identify breeds for efficient farm management.
3. Pet & Veterinary Use: Enable pet recognition in smart devices and assist vets with image-based diagnostics.
4. Surveillance Systems: Deploy in forests or rural areas to detect endangered animals and prevent poaching.
5. Mobile & Web Apps: Power educational, tourism, or social media apps for animal identification from user images.
6. Multimodal AI: Combine image, sound, and motion data for more accurate and robust animal tracking systems.



CONCLUSION

In conclusion, animal image classification using models like EfficientNetB3 offers a powerful solution for a wide range of real-world applications. From wildlife conservation and biodiversity research to smart farming and pet identification, it brings automation, accuracy, and efficiency to animal-related tasks. By leveraging deep learning, this technology not only supports scientific and environmental efforts but also enhances everyday tools and systems. As advancements in AI continue, animal image classification will play an increasingly vital role in understanding and protecting the animal kingdom.

promova



A dark blue background featuring several geometric shapes. In the upper left, there is a small cube and a sphere. On the left side, there is a large, thick, light blue ring. The text "THANK YOU" is centered in a white, serif font.

THANK YOU

TEAM 03